

LAB ASSIGNMENT-5

Overfitting and Regularization

Objective

To study overfitting in regression and classification models by analysing (i) learning behaviour across training epochs and (ii) generalization behaviour across model complexity and regularization strength.

Common Experimental Setting

The following settings must be used throughout the assignment:

- Train-test split: **70% training, 30% testing**.
- Optimization method: gradient descent (or equivalent iterative solver).
- Number of training epochs: **100 epochs (fixed)**.
- Metrics must be recorded at the end of **each epoch**.
- All plots must clearly specify the x-axis variable.

Q1. Overfitting in Non-Linear Regression

Part A: Effect of Model Complexity

Dataset Generation

Generate a synthetic dataset with at least 600 samples and five numerical input features:

$$x_1, x_2, x_3, x_4, x_5 \sim \mathcal{U}(-1, 1).$$

Define the target variable:

$$y = 3x_1^2 + 2x_2x_3 + 0.5x_4^3 - x_5 + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.2).$$

Model Training

Construct polynomial regression models with polynomial degree

$$d \in \{1, 2, 3, 4, 5, 6\}.$$

For each degree d , train a Linear Regression model using gradient descent for **100 epochs**.

Epoch-wise Evaluation

For each degree d , record at every epoch:

- Training R^2
- Testing R^2

Epoch-wise Plot

For each polynomial degree d , plot:

- x-axis: training epoch
- y-axis: R^2

Include two curves in the same plot:

- Training R^2 vs epoch
- Testing R^2 vs epoch

Parameter-wise Evaluation

After training is complete, record the final (epoch 100) values:

- Training $R^2(d)$
- Testing $R^2(d)$

Parameter-wise Plot

Plot:

- x-axis: polynomial degree d
- y-axis: final R^2 (after 100 epochs)

Each point corresponds to a separately trained model.
Explain where overfitting begins and why.

Part B: Effect of L2 Regularization

Fix the polynomial degree at $d = 6$.

Train Ridge Regression models using gradient descent for **100 epochs** with

$$\lambda \in \{0, 0.1, 0.2, \dots, 1.0\}.$$

Epoch-wise Plot

For a selected value of λ , plot training and testing R^2 versus epoch.

Parameter-wise Plot

For all values of λ , plot:

- x-axis: λ
- y-axis: final R^2 (after 100 epochs)

Explain how regularization controls overfitting.

Q2. Overfitting in Non-Linear Binary Classification

Part A: Demonstration of Overfitting

Generate a dataset with at least 500 samples:

$$x_1, x_2, x_3, x_4, x_5 \sim \mathcal{U}(-1, 1).$$

Define:

$$z = x_1^2 + x_2^2 + 0.5x_3 - x_4x_5 + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.1).$$

Assign labels:

$$y = \begin{cases} 1, & z > 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

Train Logistic Regression using gradient descent for **100 epochs**.

Epoch-wise Plot

Plot:

- x-axis: epoch
- y-axis: accuracy

Include both training and testing accuracy curves.

Part B: Effect of L2 Regularization

Train Logistic Regression with L2 regularization for

$$\lambda \in \{0, 0.1, 0.2, \dots, 1.0\}.$$

Parameter-wise Plot

Plot:

- x-axis: λ
- y-axis: final accuracy (after 100 epochs)

Explain how L2 regularization improves generalization.

Submission Guidelines

Submit:

- Source code
- All required plots
- A short report explaining observations